



Department of Computing
Web Design and Development
Business Processes - Modelling & Design

Assignment: **Conceptualising the Problem Solution**
Due: **02nd March 2026**
Group: **7**
Project Title: **Digital Examination Attendance System**

Instructions

- a) This document acts as a reference or template to aid you in the design and development of web products based on the proposed problem ideas.
- b) Fill in the respective sections with all necessary contents based on your group project ideas. Submit via Google Classroom as a soft copy and hand in a printed hard copy. A single submission is expected per group.
- c) Any form of plagiarism will attract severe penalties. **All rules of academic writing apply.**

- d) Any other document **format** other than what is contained and specified herein will **NOT** be assessed.
This cover page must not overflow to the next page.

Group Members

#	Registration Number	Name (<i>as indicated on the portal</i>)	Score	Score	Score
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Problem Definition

In many educational institutions, examination attendance is recorded manually using paper-based sheets during the exam session. Invigilators are required to move around the examination room to verify and mark each student's presence, introducing a range of operational and administrative challenges.

Project Goal/Objective

A digital attendance system is proposed in which student attendance is recorded before entering the examination room, shifting the verification process to the entry point.

Roles/Users

Student; Invigilators; Administrative / Academic Staff

Functional Requirements

- Authentication
 - o **FR1.1** An invigilator must be able to log in to the system.
 - o **FR1.2** Only authorised invigilators and administrators can access the system.

- Session Creation
 - o **FR2.1** An administrator must be able to create an exam session.

- o **FR2.2** An administrator must be able to upload or import a list of registered students for each session.
 - o **FR2.3** The system shall associate each student with a unique institutional ID.
- ID Scanning and Verification
 - o **FR3.1** An invigilator must be able to scan a student's institutional ID card at the exam entrance and at the end of the exam.
 - o **FR3.2** The system must validate the scanned ID against the session's registered student list in real time.
 - o **FR3.3** The scan results (success or failure) must be displayed on screen within 5 seconds.
 - o **FR3.4** On successful verification, the system must mark the student as present and completed with a timestamp.
 - o **FR3.5** If a certain student did not finish the exam for certain valid reasons, the system must allow invigilators to provide remarks for the individual.
- Instant Alerts
 - o **FR4.1** The system must display an on-screen notification for duplicated scan, invalid IDs and unregistered students.
- Offline Capability
 - o **FR5.1** In case of internet outage, the system must allow offline scanning and verification of students.
 - o **FR5.2** The invigilator must be able to manually search for student by name or ID number and mark attendance offline.

- o **FR5.3** The system must store offline attendance records locally.
 - o **FR5.4** The system must automatically sync offline records to the central database when internet becomes available.
- Broken ID or Scanner Failure
 - o **FR6.1** If a student's ID card is broken, lost, or unreadable, an invigilator must be able to manually search for student by name or ID number.
 - o **FR6.2** After manual verification, the invigilator must be able to mark the student as present.
 - o **FR6.3** If the scanner device fails, the invigilator must be able to use manual lookup as a fallback.
- Live Dashboard and reports
 - o **FR7.1** An administrator must be able to view a live dashboard showing total registered, present and absent.
 - o **FR7.2** An administrator must be able to view attendance for any active session at any time.
 - o **FR7.3** The system must generate per-session attendance reports including total present, total absent and timestamps.
 - o **FR7.4** The system shall produce attendance statistics (e.g., percentage present).

REST Endpoints

Module	Endpoint	Method	Description	Request Body / Parameters
Auth	/api/v1/auth/login	POST	Invigilator or admin login	{email, password ,role}
	/api/v1/auth/logout	POST	Logout user	–
	/api/v1/auth/me	GET	Get current user profile	–
Session	/api/v1/sessions	POST	Admin creates exam session	{sessionName,date, startTime,endTime,venue}
	/api/v1/sessions/{sessionId}/students/import	POST	Upload registered student list for a session	CSV file or JSON array: {studentId,fullName,InstitutionalId}
Attendance	/api/v1/scan	POST	Scan ID, validate, mark present, mark complete, grant/deny entry	{sessionId,institutionalId,s cannerDeviceId}

	/api/v1/manual/verify	POST	Search student by name / ID when scanner fails	{sessionId,searchTerm,verifiedBy}
	/api/v1/manual/attendance	POST	Mark present and complete after manual verification	{sessionId,studentId,verifiedBy,reason}
Offline	/api/v1/offline/sync	POST	Push offline records to server when internet returns	{deviceId,offlineRecords}
Dashboard	/api/v1/sessions/{sessionId}/dashboard	GET	Live view: present/absent counts and lists	Parameter: {sessionId}
	/api/v1/sessions/{sessionId}/report	GET	Generate full attendance report	Parameters: {sessionId}, Query:?format=json

